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Security Documentation Checklist and Outline

>>>> Checklist for Security Documentation <<<<

1. Dependency Downloads

- ☐ List of Components: Document all software, libraries, and dependencies.
- ☐ Version Control: Specify versions for consistency in testing.
- ☐ Licenses: Ensure all open-source or proprietary licenses are noted.

2. Implementation Details

- ☐ Local Deployment Steps: Clear steps for installing and running the software locally.
- ☐ Execution Environment: OS requirements, storage needs, and any hardware dependencies.

3. Network Security

- ☐ Proof of Isolation: Steps proving no internet access is necessary or used.
- ☐ Firewall Rules: Any configurations for local firewall or network settings.
- ☐ Testing for Isolation: Guide IT on how they can independently verify no external connectivity.

4. Data Security and Privacy

- ☐ Data Handling: Document where and how data is stored, processed, and deleted.
- ☐ Encryption (If Applicable): Any encryption in use for local data.
- ☐ Access Control: Define who has access and how access is controlled on the local machine.

5. Safety Verification Instructions

- ☐ Test Environment Setup: Replicating the local environment.



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ð Verification Checklist: Actions to verify the deployment's isolation and functionality.

ð Troubleshooting: Basic troubleshooting steps .



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Outline for Security Documentation

I. Project Overview

- Purpose of the Software: Brief summary of the project’s role in VDRS report generation.
- Deployment Scope: Local, isolated environment; not intended for network or internet use.
- Include table of contents.

II. Downloads and Dependencies

- List all software, programming languages, and libraries.

III. Installation and Local Deployment

- Step-by-Step Installation: Detailed guide on how to set up the software on a local machine.
- System Requirements: OS compatibility, minimum hardware specifications.
 - Any modern web browser
 - Windows 10 or newer (additional OS support is available upon request)
 - AMD/Intel x86 architecture PC with a minimum 16GB RAM
 - Minimum 16GB DDR4 RAM (32GB+ DDR5 preferred)
 - 100 GB local solid state storage
 - Internet connection to download the software product for the first time. Alternatively, a physical copy can be sent via USPS Certified Mail.

IV. Network and Security Details

- Local Isolation Explanation: How the software operates fully offline.
- Verification of Isolation: Steps IT can follow to check for internet access attempts or network calls.
- Firewall and Security Settings: Any recommended configurations to enhance local isolation.

V. Data Security Measures

- Data Handling Process: How data flows through the system, including storage and deletion.
- Encryption (If Used): Brief details on any encryption protocols.
- Access Control Policies: Permissions or restrictions on the software for different users.

VI. Testing and Verification for IT

- Testing Protocols: Checklist IT can follow to validate isolation and functionality.
- Download and Execution: Guide for downloading, verifying, and testing the software locally.



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- Expected Results and Troubleshooting: Expected results for verification steps, with common troubleshooting suggestions.



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AKVDRS Abstraction Tool: Security Documentation

I. Project Overview

Purpose of the Software

This software solution (AKVDRS Abstraction Tool by Tyler Systems LLC) is designed to streamline the Alaska Violent Death Reporting System (VDRS) reporting process by assisting in the generation of narrative summaries. Utilizing a fine-tuned Large Language Model (LLM), the software processes and summarizes multiple types of documents, including law enforcement reports, medical examiner reports, and other sources, to generate a cohesive draft narrative for each case. This narrative generation tool is intended to expedite the reporting process, reduce manual workload, and support Alaska VDRS in submitting comprehensive, timely, and accurate data to the National Violent Death Reporting System (NVDRS) managed by the CDC.

Deployment Scope

The software is deployed as a fully isolated, on-premises application, operating without the need for internet or network connectivity. This offline deployment ensures the highest levels of data privacy and security, in compliance with Alaska VDRS data protection requirements. By maintaining local-only operations, the software minimizes potential cybersecurity risks and meets regulatory and organizational security standards.

Target Audience and User Roles

The primary users of this software are Alaska VDRS data abstractors and epidemiologists who are responsible for reviewing and coding case information related to violent deaths. These professionals will interact with the application to:

- Upload various document types associated with each case.
- Generate draft narrative summaries based on the content of these documents.
- Review, edit, and finalize the output before submitting data to the CDC.

The software is designed to be user-friendly, requiring minimal technical expertise from the users. It integrates seamlessly into the current VDRS workflow, reducing manual work and freeing up time for staff to focus on data quality and other critical tasks.



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Key Objectives

This project is committed to achieving two primary objectives:

1. **Efficiency and Automation:** Reduce the time and resources required for Alaska VDRS personnel to compile narrative summaries by leveraging AI-driven automation.
2. **Data Security and Privacy:** Ensure that all sensitive information remains protected by keeping the software fully offline and isolated, thus preventing unauthorized access or data exposure.

These objectives align with the overarching goals of improving data quality, reducing backlogs, and supporting the Alaska VDRS team in their mission to provide timely, accurate reporting on violent deaths within the state.

Table of Contents

The documentation is organized as follows to facilitate ease of reference and support IT in verifying compliance with security and deployment standards:

1. [Project Overview \[Page 04\]](#): Background, purpose, and deployment scope of the software.
2. [Software Stack and Dependencies \[Page 06\]](#): List of all software components, versions, and licensing information.
3. [Installation and Local Deployment \[Page 09\]](#): Detailed setup instructions, system requirements, and hardware specifications.
4. [Network and Security Details \[Page 10\]](#): Overview of local-only operation, isolation verification, and firewall configurations.
5. [Data Security Measures \[Page 13\]](#): Data handling procedures, encryption practices, and access control policies.
6. [Testing and Verification for IT \[Page 15\]](#): Step-by-step testing guidelines for IT to verify software isolation and functionality.
7. [Applicable Licenses \[Page 17\]](#): Licensing details for each software component, including the LLM and its acceptable use policy.



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II. Software Stack and Dependencies

This section provides a comprehensive list of all software components, programming languages, and libraries used within the application. For each item, we outline the version, licensing information, and installation sources, ensuring transparency and ease of verification.

Template for Software Components

Each software component is documented with the following details:

Component Name: Name of the software, library, or tool.

Version: Specific version number to ensure consistency in deployment.

License: Type of license and any applicable usage terms.

Source/Installation Link: URL or method to obtain the software.

Purpose: Brief explanation of how the component is used within the system.

A. Software, Programming Languages, and Libraries

Name of Dependency	File type	Specific purpose/appl ication	Version number	Licensing details	Installation source
Meta Llam a 3.1 8B		Serves as the main large language model for narrative generation, processing and summarizing input documents to produce draft narratives for VDRS cases.	3.1	Llama 3.1 Community License by Meta Platforms, Inc.	Llama Model Downloads
PDF Parsing Utility					
R studio		IDE for R, Python, HTML, CSS, and Javascript development . Facilitates data handling and analysis	4.2 or higher	GNU General Public License (GPL)	



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		functions within the application. R is particularly used for integration with statistical models and Shiny deployment, if required.			
Github					
Microsoft Sharepoint					

1. Meta Llama 3.1 8B

- **Version:**
- **License:**
- **Source/Installation Link:**
- **Purpose:**

6. PDF Parsing Utility

- **Version:** [Specify version based on chosen library, e.g., PDFPlumber, PDF.js]
- **License:** [Specify license type, e.g., MIT, GPL]
- **Source/Installation Link:** [Library-specific download link or repository]
- **Purpose:** Extracts text and metadata from PDF documents, enabling the application to process these files as input for narrative generation.

8. GitHub

- **Version:** N/A (Web-based service)
- **License:** GitHub Terms of Service
- **Source/Installation Link:** [GitHub](#)
- **Purpose:** Used for version control and collaborative code management, storing scripts and assets relevant to the project.

9. Microsoft SharePoint

- **Version:** Cloud-based, managed by organization's Office 365 plan



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- **License:** Microsoft Enterprise License Agreement
- **Source/Installation Link:** [Microsoft Office 365](#)
- **Purpose:** Used for secure document sharing and collaboration among the development and client teams, particularly for non-code documentation and project assets.



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III. Installation and Local Deployment

Step-by-Step Installation

This section will provide a comprehensive, step-by-step guide to install and configure the software on a local machine.

- **Instructions:** Step-by-step installation instructions will be added here, covering each aspect of setting up the environment, verifying dependencies, and configuring local settings for optimal performance.
- **Additional Resources:** Links to any required installation assets, configuration files, or troubleshooting guides will also be provided in this section.

[Placeholder for detailed installation instructions.]

System Requirements

The following system requirements ensure optimal performance and compatibility:

[Explain the difference between minimum system requirements, actual recommended requirements, capabilities difference, etc.]

1. **Operating System:** Windows 10 or newer (additional OS support available upon request).
2. **Hardware Specifications:**
 - **Processor:** AMD/Intel x86 architecture PC.
 - **Memory:** Minimum 16GB DDR4 RAM (32GB+ DDR5 preferred for enhanced performance).
 - **Storage:** 100 GB local solid-state storage.
3. **Web Browser:** Any modern web browser to access the application's interface.
4. **Internet Connection (Initial Setup Only):** An internet connection is required solely for the initial download of the software package.
 - **Alternative:** A physical copy of the software can be sent via USPS Certified Mail for installations where network access is restricted or unavailable.

[Further customization and requirements will be specified once installation instructions are finalized.]



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IV. Network and Security Details

Local Isolation Explanation

The Tyler Systems LLC solution for Alaska VDRS narrative generation is engineered to operate exclusively in a local environment, ensuring complete isolation from the internet. This design choice supports Alaska VDRS's requirement for a secure, self-contained system, minimizing risks related to data breaches, unauthorized access, or cybersecurity vulnerabilities.

Definition of an "external API"

[Insert a clear, readable definition here]

Operational Scope:

- The tool functions as a standalone application, meaning it does not rely on any online services or external APIs. All data processing occurs locally, which ensures that sensitive information remains within the controlled environment of the on-premises machine.
- This offline functionality safeguards sensitive Alaska VDRS data from exposure beyond the local network, reinforcing data security and compliance with data privacy regulations.

Data Privacy and Security Consideration:

This approach aligns with data privacy regulations and adheres to Alaska VDRS's data security policies, ensuring a controlled and secure environment for handling sensitive incident data.

Verification of Isolation

To verify the application's security, the following procedural guidance enables IT staff to confirm that the application is fully disconnected from any internet access.

1. Network Monitoring

- **Procedure:** IT staff can use network monitoring tools (e.g., Wireshark, NetFlow) to monitor outgoing and incoming network traffic when the software is in operation.
- **Expected Result:** No outbound or inbound network connections should be detected during regular operation, affirming that the application does not attempt any external communications.

2. Firewall Configuration and Verification

- **Procedure:** Confirm that firewall settings restrict all outgoing connections from the machine running the software. IT staff should configure the firewall to deny all external access attempts from the application's environment.



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- **Expected Result:** The firewall logs should show no attempts to establish a connection to external IP addresses, providing evidence of isolation.

3. Airplane Mode Testing

- **Procedure:** Activate airplane mode or physically disconnect network cables, then launch and operate the software.
- **Expected Result:** The software should run without any interruption or functionality loss, demonstrating its independence from network connectivity.

4. Review of Software Logs

- **Procedure:** Review log files generated by the software during operation to ensure they do not contain entries related to network access or connection attempts.
- **Expected Result:** Logs should reflect only local operations without any recorded attempts to reach external servers or IP addresses.

5. Virtual Machine (VM) Isolation Test

- **Procedure:** Run the software in a virtual machine (VM) with all network adapters disabled.
- **Expected Result:** The software should perform all functions as expected within the VM environment, confirming that it does not depend on external connections.

Optional Firewall and Security Settings

To further reinforce the security of the deployment, the following firewall and security configurations are recommended:

1. Default-Deny Outbound Rules

- **Configuration:** Establish a default-deny rule for all outbound connections on the machine hosting the application. This ensures that no unauthorized traffic can exit the system, even if an external connection is attempted inadvertently.
- **Logging:** Enable logging for blocked connection attempts to maintain visibility into any unexpected or unauthorized network activity.

2. Port Restrictions

- **Configuration:** Confirm that the software does not require any open ports for communication, reducing potential vulnerabilities associated with exposed ports. If any local services are required, restrict ports to only those essential for operation and monitor these for security purposes.



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3. Enhanced System Hardening Recommendations:

- Disable any unnecessary services or open ports on the system to reduce the attack surface.
- Regularly update the operating system and other critical software to close security vulnerabilities and enhance resilience against potential threats.

Additional Isolation Assurance

The Tyler Systems LLC team is committed to supporting any additional verification processes required by Alaska VDRS IT staff or its affiliates. This includes collaborating on security assessments, isolation testing, and reviewing logs or configurations to further substantiate the solution's security posture.

Documentation and technical support are available for setting up and running these verification steps, ensuring that the Alaska VDRS IT team has full confidence in the system's isolation and data protection capabilities.



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V. Data Security Measures

This section outlines the protocols and security measures implemented to safeguard sensitive data within the Alaska VDRS narrative generation application. The focus is on ensuring data privacy, secure storage, controlled access, and systematic deletion.

Data Handling Process

- **Overview:** This application handles sensitive data related to violent death reporting. All data remains on local storage to ensure security and compliance with privacy requirements.
- **Data Flow:** Data is processed in a local environment, with inputs restricted to PDF, and DOCX files. The application processes each document type, extracting information needed to generate narrative summaries for VDRS cases.
- **Data Storage:** Temporary files are stored locally during active sessions only. Once processing is complete, all temporary data is securely deleted to prevent unauthorized access.
- **Deletion Protocol:**
 - Data is automatically purged after the completion of each session.
 - Users may also trigger manual deletion of files or clear cached data, ensuring sensitive information is not retained beyond the active session.

Encryption

- **Local Encryption (If Implemented):** All data stored temporarily during a session is encrypted using industry-standard encryption protocols to prevent unauthorized access.
- **Encryption Standards:** Encryption protocols such as AES-256 are preferred to secure data in transit (internally within the local system) and at rest (temporary storage).
- **Future Encryption Needs:** For future updates, considerations will include enhancing data encryption for cached data and implementing advanced encryption methods if required by Alaska VDRS security standards.

Access Control Policies

- **User Access Restrictions:** Access to the application is limited to authorized Alaska VDRS personnel only, following organizational policies for user authentication.
- **Role-Based Access Control:** If multiple roles are introduced, role-based access may be implemented to restrict access to specific features or data based on the user's role within the organization.
- **Login and Authentication (If Applicable):** Login credentials or system-based authentication may be introduced, with plans to store credentials locally and securely within the isolated environment.



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- **Physical Security:** The system is designed to run on on-premises hardware, ensuring physical security controls are in place in line with Alaska VDRS protocols. Access to machines running the software should be restricted to authorized personnel.

Regular Audits and Compliance

- **Audit Logs:** Audit logs may be implemented in future updates to track system access and operations. These logs would be stored locally and available only to authorized personnel for compliance verification.
- **Compliance with Data Security Policies:** The data security measures align with both Alaska VDRS and CDC requirements for safeguarding sensitive data, ensuring all data handling and storage protocols meet relevant regulatory standards.
- **Periodic Review:** Security policies and data handling processes will be periodically reviewed to ensure continued compliance and to incorporate new security measures as needed.



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VI. Testing and Verification

This section outlines the procedures that IT staff can follow to validate the software's isolation from external networks, functionality, and security configuration. Following these protocols will ensure compliance with Alaska VDRS data security standards and provide confidence in the application's offline operation.

Testing Protocols

- **Objective:** Confirm that the software operates fully offline, with no dependencies on external networks or cloud-based resources, and that it performs as expected in the local environment.
- **Recommended Tools:** IT can use standard network monitoring tools (e.g., Wireshark) and system auditing tools to verify isolation and local performance.

Step-by-Step Verification Checklist

1. **Isolation Test**
 - **Procedure:** Disconnect the machine from the network and operate the software in offline mode.
 - **Expected Result:** The software should run without errors or interruptions, confirming it does not rely on any internet connection.
2. **Network Traffic Analysis**
 - **Procedure:** Use a network monitoring tool to observe network activity during software operation.
 - **Expected Result:** No network traffic should be generated by the software. This will validate that it does not initiate any external communications.
3. **Firewall Configuration Validation**
 - **Procedure:** Verify that outbound network traffic is blocked by default for this application's environment. Review firewall logs for any attempts to access external networks.
 - **Expected Result:** Logs should show no attempts by the software to connect externally, confirming isolation.
4. **Local Storage Verification**
 - **Procedure:** Monitor local storage use during software operation to ensure sensitive data is only temporarily stored and that data is securely deleted after processing.
 - **Expected Result:** Local files related to the application's operation should appear only during active sessions and be deleted upon completion.



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5. Log File Review

- **Procedure:** Review log files generated by the application to verify they contain no entries indicating network access attempts or external server connections.
- **Expected Result:** Logs should reflect purely local operations, with no references to external network activity.

Download and Execution Verification

- **Objective:** Confirm the integrity of the software package and verify that it can be downloaded and installed in compliance with Alaska VDRS IT requirements.
- **Procedure:** Download the software package from a designated secure source or receive a physical copy. Follow standard installation procedures, then run the software in offline mode.
- **Expected Result:** The software should install and execute without network connectivity, confirming independence from internet resources.

Expected Results and Troubleshooting

Expected Behavior: Under standard operations, the application should function without internet access, generating no network traffic and temporarily storing data locally only when necessary.

Troubleshooting Guide:

- **Network Access Detected:** If network activity is detected, ensure firewall settings block all outbound traffic. Restart and monitor the software to confirm isolation.
- **Data Persistence Issues:** If temporary files are not deleted post-session, review local storage configurations and settings for automatic cleanup. Manually clear temporary storage if needed and notify support.
- **Software Execution Errors:** If errors occur while running offline, verify that the installation process was completed correctly and check system compatibility with the software's requirements.
- **Contact Technical Support Directly:** Speak with a system administrator or reach out to support@tylersystems.com.



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Applicable Licenses

Llama 3.1 8B Applicable Licenses

Summary

A detailed FAQ is available directly from Meta, here: <https://www.llama.com/faq/>

Llama models are broadly available to developers and licensees through a variety of hosting providers and on the Meta website. The Llama models are licensed under the applicable Llama Community License Agreement and accompanying Acceptable Use Policy, which provides a permissive license to the models along with certain restrictions to help ensure that the models are used responsibly.

Llama 3.1 8B Model Card (Description)

Definition of a Model Card

[Complete this section]

Model Information

The Meta Llama 3.1 collection of multilingual large language models (LLMs) is a collection of pretrained and instruction tuned generative models in 8B, 70B and 405B sizes (text in/text out). The Llama 3.1 instruction tuned text only models (8B, 70B, 405B) are optimized for multilingual dialogue use cases and outperform many of the available open source and closed chat models on common industry benchmarks.

Model developer: Meta

Model Architecture: Llama 3.1 is an auto-regressive language model that uses an optimized transformer architecture. The tuned versions use supervised fine-tuning (SFT) and reinforcement learning with human feedback (RLHF) to align with human preferences for helpfulness and safety.

Supported languages: English, German, French, Italian, Portuguese, Hindi, Spanish, and Thai.

Llama 3.1 family of models. Token counts refer to pretraining data only. All model versions use Grouped-Query Attention (GQA) for improved inference scalability.

Model Release Date: July 23, 2024.

Status: This is a static model trained on an offline dataset. Future versions of the tuned models will be released as we improve model safety with community feedback.



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License: A custom commercial license, the Llama 3.1 Community License, is available at:

https://github.com/meta-llama/llama-models/blob/main/models/llama3_1/LICENSE

Feedback: Instructions on how to provide feedback or comments on the model can be found in the Llama Models [README](#). For more technical information about generation parameters and recipes for how to use Llama 3.1 in applications, please go [here](#).

Intended Use

Intended Use Cases Llama 3.1 is intended for commercial and research use in multiple languages. Instruction tuned text only models are intended for assistant-like chat, whereas pretrained models can be adapted for a variety of natural language generation tasks. The Llama 3.1 model collection also supports the ability to leverage the outputs of its models to improve other models including synthetic data generation and distillation. The Llama 3.1 Community License allows for these use cases.

Out-of-scope Use in any manner that violates applicable laws or regulations (including trade compliance laws). Use in any other way that is prohibited by the Acceptable Use Policy and Llama 3.1 Community License. Use in languages beyond those explicitly referenced as supported in this model card.

Note: Llama 3.1 has been trained on a broader collection of languages than the 8 supported languages. Developers may fine-tune Llama 3.1 models for languages beyond the 8 supported languages provided they comply with the Llama 3.1 Community License and the Acceptable Use Policy and in such cases are responsible for ensuring that any uses of Llama 3.1 in additional languages is done in a safe and responsible manner.

Hardware and Software

Training Factors We used custom training libraries, Meta’s custom built GPU cluster, and production infrastructure for pretraining. Fine-tuning, annotation, and evaluation were also performed on production infrastructure.

Training Energy Use Training utilized a cumulative of **39.3M** GPU hours of computation on H100-80GB (TDP of 700W) type hardware, per the table below. Training time is the total GPU time required for training each model and power consumption is the peak power capacity per GPU device used, adjusted for power usage efficiency.

Training Greenhouse Gas Emissions Estimated total location-based greenhouse gas emissions were **11,390** tons CO₂eq for training. Since 2020, Meta has maintained net zero greenhouse gas emissions in its global operations and matched 100% of its electricity use with renewable energy, therefore the total market-based greenhouse gas emissions for training were 0 tons CO₂eq.



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The methodology used to determine training energy use and greenhouse gas emissions can be found [here](#). Since Meta is openly releasing these models, the training energy use and greenhouse gas emissions will not be incurred by others.

Training Data

Overview: Llama 3.1 was pretrained on ~15 trillion tokens of data from publicly available sources. The fine-tuning data includes publicly available instruction datasets, as well as over 25M synthetically generated examples.

Data Freshness: The pretraining data has a cutoff of December 2023.

Benchmark scores

In this section, we report the results for Llama 3.1 models on standard automatic benchmarks. For all the evaluations, we use our internal evaluations library. Details of our evals can be found [here](#). We are also releasing the raw data generated as part of our evals which can be found [here](#) in the dataset sections.

Responsibility & Safety

As part of our Responsible release approach, we followed a three-pronged strategy to managing trust & safety risks:

- Enable developers to deploy helpful, safe and flexible experiences for their target audience and for the use cases supported by Llama.
- Protect developers against adversarial users aiming to exploit Llama capabilities to potentially cause harm.
- Provide protections for the community to help prevent the misuse of our models.

Responsible deployment

Llama is a foundational technology designed to be used in a variety of use cases, examples on how Meta's Llama models have been responsibly deployed can be found in our [Community Stories webpage](#). Our approach is to build the most helpful models enabling the world to benefit from the technology power, by aligning our model safety for the generic use cases addressing a standard set of harms. Developers are then in the driver seat to tailor safety for their use case, defining their own policy and deploying the models with the necessary safeguards in their Llama systems. Llama 3.1 was developed following the best practices outlined in our Responsible Use Guide, you can refer to the [Responsible Use Guide](#) to learn more.

Llama 3.1 instruct

Our main objectives for conducting safety fine-tuning are to provide the research community with a valuable resource for studying the robustness of safety fine-tuning, as well as to offer developers a readily available, safe, and powerful model for various applications to reduce the



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developer workload to deploy safe AI systems. For more details on the safety mitigations implemented please read the Llama 3 paper.

Fine-tuning data

We employ a multi-faceted approach to data collection, combining human-generated data from our vendors with synthetic data to mitigate potential safety risks. We've developed many large language model (LLM)-based classifiers that enable us to thoughtfully select high-quality prompts and responses, enhancing data quality control.

Refusals and Tone

Building on the work we started with Llama 3, we put a great emphasis on model refusals to benign prompts as well as refusal tone. We included both borderline and adversarial prompts in our safety data strategy, and modified our safety data responses to follow tone guidelines.

Llama 3.1 systems

Large language models, including Llama 3.1, are not designed to be deployed in isolation but instead should be deployed as part of an overall AI system with additional safety guardrails as required. Developers are expected to deploy system safeguards when building agentic systems. Safeguards are key to achieve the right helpfulness-safety alignment as well as mitigating safety and security risks inherent to the system and any integration of the model or system with external tools.

As part of our responsible release approach, we provide the community with [safeguards](#) that developers should deploy with Llama models or other LLMs, including Llama Guard 3, Prompt Guard and Code Shield. All our [reference implementations](#) demos contain these safeguards by default so developers can benefit from system-level safety out-of-the-box.

New capabilities

Note that this release introduces new capabilities, including a longer context window, multilingual inputs and outputs and possible integrations by developers with third party tools. Building with these new capabilities requires specific considerations in addition to the best practices that generally apply across all Generative AI use cases.

Tool-use: Just like in standard software development, developers are responsible for the integration of the LLM with the tools and services of their choice. They should define a clear policy for their use case and assess the integrity of the third party services they use to be aware of the safety and security limitations when using this capability. Refer to the Responsible Use Guide for best practices on the safe deployment of the third party safeguards.



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Multilinguality: Llama 3.1 supports 7 languages in addition to English: French, German, Hindi, Italian, Portuguese, Spanish, and Thai. Llama may be able to output text in other languages than those that meet performance thresholds for safety and helpfulness. We strongly discourage developers from using this model to converse in non-supported languages without implementing finetuning and system controls in alignment with their policies and the best practices shared in the Responsible Use Guide.

Evaluations

We evaluated Llama models for common use cases as well as specific capabilities. Common use cases evaluations measure safety risks of systems for most commonly built applications including chat bot, coding assistant, tool calls. We built dedicated, adversarial evaluation datasets and evaluated systems composed of Llama models and Llama Guard 3 to filter input prompt and output response. It is important to evaluate applications in context, and we recommend building dedicated evaluation dataset for your use case. Prompt Guard and Code Shield are also available if relevant to the application.

Capability evaluations measure vulnerabilities of Llama models inherent to specific capabilities, for which were crafted dedicated benchmarks including long context, multilingual, tools calls, coding or memorization.

Red teaming

For both scenarios, we conducted recurring red teaming exercises with the goal of discovering risks via adversarial prompting and we used the learnings to improve our benchmarks and safety tuning datasets.

We partnered early with subject-matter experts in critical risk areas to understand the nature of these real-world harms and how such models may lead to unintended harm for society. Based on these conversations, we derived a set of adversarial goals for the red team to attempt to achieve, such as extracting harmful information or reprogramming the model to act in a potentially harmful capacity. The red team consisted of experts in cybersecurity, adversarial machine learning, responsible AI, and integrity in addition to multilingual content specialists with background in integrity issues in specific geographic markets.

Critical and other risks

We specifically focused our efforts on mitigating the following critical risk areas:

1. CBRNE (Chemical, Biological, Radiological, Nuclear, and Explosive materials) helpfulness

To assess risks related to proliferation of chemical and biological weapons, we performed uplift testing designed to assess whether use of Llama 3.1 models could meaningfully increase the capabilities of malicious actors to plan or carry out attacks using these types of weapons.

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2. Child Safety

Child Safety risk assessments were conducted using a team of experts, to assess the model's capability to produce outputs that could result in Child Safety risks and inform on any necessary and appropriate risk mitigations via fine tuning. We leveraged those expert red teaming sessions to expand the coverage of our evaluation benchmarks through Llama 3 model development. For Llama 3, we conducted new in-depth sessions using objective based methodologies to assess the model risks along multiple attack vectors including the additional languages Llama 3 is trained on. We also partnered with content specialists to perform red teaming exercises assessing potentially violating content while taking account of market specific nuances or experiences.

3. Cyber attack enablement

Our cyber attack uplift study investigated whether LLMs can enhance human capabilities in hacking tasks, both in terms of skill level and speed.

Our attack automation study focused on evaluating the capabilities of LLMs when used as autonomous agents in cyber offensive operations, specifically in the context of ransomware attacks. This evaluation was distinct from previous studies that considered LLMs as interactive assistants. The primary objective was to assess whether these models could effectively function as independent agents in executing complex cyber-attacks without human intervention.

Our study of Llama 3.1 405B's social engineering uplift for cyber attackers was conducted to assess the effectiveness of AI models in aiding cyber threat actors in spear phishing campaigns. Please read our Llama 3.1 Cyber security whitepaper to learn more.

Community

Generative AI safety requires expertise and tooling, and we believe in the strength of the open community to accelerate its progress. We are active members of open consortiums, including the AI Alliance, Partnership on AI and MLCommons, actively contributing to safety standardization and transparency. We encourage the community to adopt taxonomies like the MLCommons Proof of Concept evaluation to facilitate collaboration and transparency on safety and content evaluations. Our Purple Llama tools are open sourced for the community to use and widely distributed across ecosystem partners including cloud service providers. We encourage community contributions to our [Github repository](#).

We also set up the [Llama Impact Grants](#) program to identify and support the most compelling applications of Meta's Llama model for societal benefit across three categories: education, climate and open innovation. The 20 finalists from the hundreds of applications can be found [here](#).



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Finally, we put in place a set of resources including an [output reporting mechanism](#) and [bug bounty program](#) to continuously improve the Llama technology with the help of the community.

Ethical Considerations and Limitations

The core values of Llama 3.1 are openness, inclusivity and helpfulness. It is meant to serve everyone, and to work for a wide range of use cases. It is thus designed to be accessible to people across many different backgrounds, experiences and perspectives. Llama 3.1 addresses users and their needs as they are, without insertion unnecessary judgment or normativity, while reflecting the understanding that even content that may appear problematic in some cases can serve valuable purposes in others. It respects the dignity and autonomy of all users, especially in terms of the values of free thought and expression that power innovation and progress.

But Llama 3.1 is a new technology, and like any new technology, there are risks associated with its use. Testing conducted to date has not covered, nor could it cover, all scenarios. For these reasons, as with all LLMs, Llama 3.1's potential outputs cannot be predicted in advance, and the model may in some instances produce inaccurate, biased or other objectionable responses to user prompts. Therefore, before deploying any applications of Llama 3.1 models, developers should perform safety testing and tuning tailored to their specific applications of the model. Please refer to available resources including our [Responsible Use Guide](#), [Trust and Safety](#) solutions, and other [resources](#) to learn more about responsible development.



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Llama 3.1 Version Release Date: July 23, 2024

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6. Term and Termination. The term of this Agreement will commence upon your acceptance of this Agreement or access to the Llama Materials and will continue in full force and effect until terminated in accordance with the terms and conditions herein. Meta may terminate this Agreement if you are in breach of any term or condition of this Agreement. Upon termination of this Agreement, you shall delete and cease use of the Llama Materials. Sections 3, 4 and 7 shall survive the termination of this Agreement.

7. Governing Law and Jurisdiction. This Agreement will be governed and construed under the laws of the State of California without regard to choice of law principles, and the UN Convention on Contracts for the International Sale of Goods does not apply to this Agreement. The courts of California shall have exclusive jurisdiction of any dispute arising out of this Agreement.



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Llama 3.1 Acceptable Use Policy

Meta is committed to promoting safe and fair use of its tools and features, including Llama 3.1. If you access or use Llama 3.1, you agree to this Acceptable Use Policy (“**Policy**”). The most recent copy of this policy can be found at https://llama.meta.com/llama3_1/use-policy.

Prohibited Uses

We want everyone to use Llama 3.1 safely and responsibly. You agree you will not use, or allow others to use, Llama 3.1 to:

- Violate the law or others’ rights, including to:
 - Engage in, promote, generate, contribute to, encourage, plan, incite, or further illegal or unlawful activity or content, such as:
 - Violence or terrorism
 - Exploitation or harm to children, including the solicitation, creation, acquisition, or dissemination of child exploitative content or failure to report Child Sexual Abuse Material
 - Human trafficking, exploitation, and sexual violence
 - The illegal distribution of information or materials to minors, including obscene materials, or failure to employ legally required age-gating in connection with such information or materials.
 - Sexual solicitation
 - Any other criminal activity
 - Engage in, promote, incite, or facilitate the harassment, abuse, threatening, or bullying of individuals or groups of individuals
 - Engage in, promote, incite, or facilitate discrimination or other unlawful or harmful conduct in the provision of employment, employment benefits, credit, housing, other economic benefits, or other essential goods and services
 - Engage in the unauthorized or unlicensed practice of any profession including, but not limited to, financial, legal, medical/health, or related professional practices
 - Collect, process, disclose, generate, or infer health, demographic, or other sensitive personal or private information about individuals without rights and consents required by applicable laws
 - Engage in or facilitate any action or generate any content that infringes, misappropriates, or otherwise violates any third-party rights, including the outputs or results of any products or services using the Llama Materials
 - Create, generate, or facilitate the creation of malicious code, malware, computer viruses or do anything else that could disable, overburden, interfere with or impair the proper working, integrity, operation or appearance of a website or computer system



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- Engage in, promote, incite, facilitate, or assist in the planning or development of activities that present a risk of death or bodily harm to individuals, including use of Llama 3.1 related to the following:
 - Military, warfare, nuclear industries or applications, espionage, use for materials or activities that are subject to the International Traffic Arms Regulations (ITAR) maintained by the United States Department of State
 - Guns and illegal weapons (including weapon development)
 - Illegal drugs and regulated/controlled substances
 - Operation of critical infrastructure, transportation technologies, or heavy machinery
 - Self-harm or harm to others, including suicide, cutting, and eating disorders
 - Any content intended to incite or promote violence, abuse, or any infliction of bodily harm to an individual
- Intentionally deceive or mislead others, including use of Llama 3.1 related to the following:
 - Generating, promoting, or furthering fraud or the creation or promotion of disinformation
 - Generating, promoting, or furthering defamatory content, including the creation of defamatory statements, images, or other content
 - Generating, promoting, or further distributing spam
 - Impersonating another individual without consent, authorization, or legal right
 - Representing that the use of Llama 3.1 or outputs are human-generated
 - Generating or facilitating false online engagement, including fake reviews and other means of fake online engagement
- Fail to appropriately disclose to end users any known dangers of your AI system

Please report any violation of this Policy, software “bug,” or other problems that could lead to a violation of this Policy through one of the following means:

Reporting issues with the model: <https://github.com/meta-llama/llama-models/issues>

Reporting risky content generated by the model:
developers.facebook.com/llama_output_feedback

Reporting bugs and security concerns: facebook.com/whitehat/info

Reporting violations of the Acceptable Use Policy or unlicensed uses of Llama 3.1:
LlamaUseReport@meta.com



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